

G'MIC: A Pleasant Image-Processing Framework

JAI SACHITH PAUL

In the last issue of EFY Plus, we discussed Clmg library for image processing. Let us now take a look at GREYC's Magic for Image Computing (G'MIC) tool, which is based on the latest developments of Clmg.

G'MIC is a pleasant yet powerful image-processing framework that comes with an extensive package of top-quality effects and predefined filters. The latest release of the software version 1.6.5.0 is included in this month's DVD of EFY Plus. The software is distributed under CEA CNRS INRIA Logiciel Libre (CeCILL) licence, which is a French GPL-compatible licence.

Why G'MIC

Even though the purpose of G'MIC is to facilitate experts, there is some-

thing for everyone in it. As a beginner, one can easily use this tool for functions like mirroring, rotating, sharpening, resizing, creating animations using various warps and deformations. An expert can use this tool to work with videos, for processing webcam videos in real-time, to have multi-dimensional plots, for creating 3D models and logos, and so on. Manipulations and interactions on the image can be done either grouped or focused on specific items.

Simplicity. A simple instruction, `gmic pic1.jpg pic2.jpg pic3.jpg`, allows the user to view three pictures in a sequence.

Conversion of an image from one format to another is possible by yet another single-line instruction, `gmic ThisPicture.png Output.jpg`.

A simple command, `gmic image.jpg -old_photo`, allows the user to create an old-photo effect for an image.

Flexibility. The tool gives the user the freedom to choose from a range of image-processing options. For example, for blurring an image, the user is given an option to select filter type, border setting,

various blur types (angular, radial, linear and selective), direction ($x/xy/xyz/y/z$) and more.

Extensible. The software allows the user to have a customised library of commands that could be understood by the interpreter. Thus, an expert user can customise the library in order to adapt to his or her requirements.

Portability. Developers of this software had portability as one of their major design considerations. G'MIC runs well on different platforms such as Windows, Unix and Mac. Use of OpenMP specifications takes advantage of multiple cores for speeding up the computation of image-processing operations while parallel-processing.

Motivation behind the development

As mentioned earlier, the same team who developed Clmg library developed G'MIC framework. There were some observations that led to the development of the new framework.

For working on Clmg, basic knowledge of C++ is necessary. We cannot expect people like doctors, who are interested in image processing, to have literacy on the programming language.

Some recurring basic operations

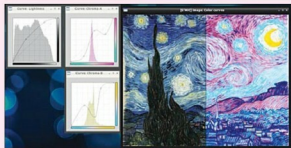


Fig. 1: Colour transformation using interactive colour curves



Fig. 2: Comic colourisation by placing coloured keypoints and running Colorization algorithm for interpolating the colours (Image courtesy: David Revoy website)

Software overview

Name of the tool: G'MIC (GREYC's Magic for Image Computing)
Application: Image processing
Main developer: David Tschumperlé, GREYC Lab, Caen France
Main coding languages: C and C++
Licence: CeCILL-2.0 (GPL-compatible)
Latest release: Version 1.6.5.0